

AI Fashion Recommendation Systems and Virtual Try-On Research Documentation

Raritone Summer Internship 2026

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Declaration

I hereby declare that this documentation contains the research, analysis, findings, workflows, and learning outcomes completed during my AI/ML Internship at Raritone Summer Internship 2026. The information presented in this document is based on the research conducted on fashion recommendation systems, virtual try-on technologies, fashion datasets, and Artificial Intelligence applications in the fashion industry.

This documentation summarizes the work completed throughout the internship and highlights the knowledge gained through studying various recommendation techniques, virtual try-on models, and emerging AI technologies.

Introduction

The fashion industry has undergone significant digital transformation in recent years, with Artificial Intelligence becoming one of the key technologies driving innovation. Modern consumers expect personalized shopping experiences, accurate product recommendations, and interactive ways to visualize products before making purchasing decisions.

Online fashion platforms often face challenges such as helping customers discover relevant products and reducing uncertainty during online purchases. To address these challenges, companies increasingly rely on recommendation systems and virtual try-on technologies powered by Artificial Intelligence.

Recommendation systems analyze customer preferences, browsing history, and shopping behavior to provide personalized suggestions. Virtual try-on technologies allow customers to visualize garments digitally before purchasing, helping improve confidence and reduce product returns.

This project focuses on understanding how these technologies work, the datasets used to train them, and the different AI models that contribute to improving online fashion experiences. The research explores recommendation algorithms, virtual try-on frameworks, and future opportunities for AI-driven fashion solutions.

Overview of AI in Fashion

Artificial Intelligence has become an important component of modern fashion technology. Fashion companies use AI to improve customer experiences, optimize business operations, and deliver highly personalized services.

One of the most common applications of AI in fashion is product recommendation. By analyzing customer interactions and purchase patterns, AI systems can identify preferences and recommend products that are most likely to interest individual users.

AI is also widely used for computer vision tasks such as clothing recognition, image classification, garment segmentation, and virtual try-on applications. These technologies enable customers to interact with products in new and innovative ways.

Another important application is trend prediction. AI systems can analyze social media content, fashion blogs, customer behavior, and market trends to predict upcoming fashion preferences. This information helps businesses make informed decisions regarding inventory management and product development.

The integration of Artificial Intelligence into fashion has improved personalization, customer satisfaction, and operational efficiency, making it one of the fastest-growing areas of AI adoption.

Fashion Recommendation Systems

Fashion recommendation systems are designed to help users discover products that match their interests, preferences, and shopping behavior. These systems play a critical role in modern e-commerce platforms because they improve user engagement and increase customer satisfaction.

Recommendation systems work by analyzing various types of information, including user interactions, purchase history, product attributes, ratings, reviews, and browsing behavior. Based on this information, the system generates personalized suggestions for each user.

The primary goal of recommendation systems is to reduce the effort required for customers to find products they are likely to purchase. By presenting relevant suggestions, these systems create a more personalized shopping experience and encourage user engagement.

Fashion recommendation systems are particularly important because fashion choices are highly personal and influenced by factors such as style preferences, colors, occasions, seasonal trends, and individual tastes. AI-powered recommendation engines help manage this complexity by delivering recommendations tailored to each user.

Modern recommendation systems use a combination of traditional algorithms and advanced machine learning techniques to continuously improve recommendation quality and adapt to changing user preferences.

Content-Based Filtering

Content-Based Filtering is one of the most widely used recommendation techniques in modern recommendation systems. This approach recommends products based on a user's previous interactions, preferences, and interests. Instead of relying on the behavior of other users, the system focuses on the characteristics of products that the user has already liked or purchased.

In the fashion industry, Content-Based Filtering analyzes product attributes such as clothing category, color, style, brand, fabric type, and design patterns. For example, if a user frequently interacts with casual blue denim jackets, the recommendation system is likely to suggest similar products that share the same characteristics.

The primary advantage of Content-Based Filtering is its ability to provide highly personalized recommendations. Since the recommendations are based on an individual user's preferences, the system can generate relevant suggestions even when there is limited information about other users.

However, this approach also has limitations. The system may repeatedly recommend similar products and struggle to introduce new styles or categories that the user has not previously explored. This can reduce recommendation diversity over time.

Despite these limitations, Content-Based Filtering remains an important component of many fashion recommendation systems due to its simplicity and effectiveness in understanding user preferences.

Collaborative Filtering

Collaborative Filtering is another popular recommendation technique that generates suggestions based on the behavior and preferences of multiple users. Instead of focusing only on product attributes, this approach identifies relationships between users and their interactions with products.

The fundamental idea behind Collaborative Filtering is that users with similar interests are likely to enjoy similar products. For example, if two users have purchased or liked many of the same clothing items, the system assumes they share similar preferences and may recommend products liked by one user to the other.

Collaborative Filtering can be divided into User-Based Collaborative Filtering and Item-Based Collaborative Filtering. User-Based Collaborative Filtering identifies users with similar preferences, while Item-Based Collaborative Filtering identifies products that are frequently liked together.

One of the major strengths of Collaborative Filtering is its ability to discover products that a user may not have considered previously. This helps increase recommendation diversity and introduces users to new styles and trends.

However, Collaborative Filtering depends heavily on historical user interaction data. When a new user joins the platform or a new product is introduced, the system may have insufficient information to generate accurate recommendations. This challenge is commonly known as the cold-start problem.

Despite these challenges, Collaborative Filtering remains one of the most successful recommendation techniques used in e-commerce and fashion applications.

Hybrid Recommendation Systems

Hybrid Recommendation Systems combine multiple recommendation techniques to overcome the limitations of individual approaches and improve overall recommendation quality.

In many modern fashion platforms, Content-Based Filtering and Collaborative Filtering are integrated into a single recommendation framework. By combining these techniques, the system can generate more accurate, diverse, and personalized recommendations.

For example, Content-Based Filtering may recommend products similar to those previously viewed by a user, while Collaborative Filtering introduces new products based on the preferences of similar users. The combination of these approaches creates a more balanced recommendation experience.

Hybrid systems help address common issues such as limited recommendation diversity, over-specialization, and cold-start problems. They can also improve recommendation accuracy by utilizing multiple sources of information simultaneously.

Many leading fashion and e-commerce platforms use Hybrid Recommendation Systems because they offer better performance and adaptability compared to single-method approaches. As recommendation technology continues to evolve, hybrid systems are becoming increasingly important in delivering personalized shopping experiences.

Deep Learning-Based Recommendations

With the advancement of Artificial Intelligence, Deep Learning has become a powerful tool for building intelligent recommendation systems. Deep Learning-Based Recommendation Systems use neural networks to analyze large volumes of user and product data and identify complex relationships that traditional methods may overlook.

These systems can learn user preferences by analyzing browsing history, purchase behavior, product interactions, and fashion trends. Unlike traditional recommendation

techniques, deep learning models can automatically extract meaningful patterns from large datasets without requiring extensive manual feature engineering.

In the fashion industry, deep learning models can analyze clothing images, textual descriptions, customer reviews, and user interactions simultaneously. This allows recommendation systems to better understand customer preferences and generate highly personalized suggestions.

One of the biggest advantages of deep learning-based recommendations is their ability to adapt to changing user behavior. As more data becomes available, the models continuously learn and improve recommendation quality.

However, deep learning systems require large datasets, significant computational resources, and complex model training procedures. Despite these challenges, they have become an essential component of modern recommendation platforms due to their superior performance and scalability.

Fashion Datasets Overview

Datasets are one of the most important components of Artificial Intelligence systems. In the fashion industry, datasets provide the information required to train, validate, and evaluate recommendation systems and virtual try-on models.

Fashion datasets typically contain clothing images, garment annotations, category labels, human pose information, segmentation masks, and product descriptions. These datasets help AI models learn the visual and structural characteristics of garments.

High-quality datasets play a critical role in improving model performance. The quality, diversity, and size of a dataset directly influence the accuracy of recommendation systems and virtual try-on technologies.

Researchers and developers use fashion datasets to train machine learning models for tasks such as clothing classification, outfit recommendation, image generation, virtual try-on, garment segmentation, and trend analysis.

Several public datasets have become widely used in fashion AI research, including DeepFashion, DeepFashion2, FashionGen, and DressCode. These datasets provide valuable resources for developing and evaluating modern fashion AI solutions.

DeepFashion Dataset

DeepFashion is one of the most widely used datasets in fashion-related Artificial Intelligence research. Developed to support computer vision tasks in the fashion industry, the dataset contains a large collection of clothing images along with detailed annotations and attribute information.

The dataset includes thousands of fashion images representing various clothing categories such as dresses, shirts, pants, jackets, skirts, and accessories. Each image contains valuable information that helps AI models understand garment characteristics and visual features.

DeepFashion is commonly used for tasks such as clothing classification, attribute prediction, image retrieval, outfit recommendation, and fashion recognition. Researchers utilize this dataset to train machine learning models that can identify and analyze different fashion products.

One of the major strengths of DeepFashion is the diversity of clothing styles and categories available within the dataset. This diversity enables AI models to learn from a wide range of fashion products and improves their ability to generalize to real-world scenarios.

The dataset has played a significant role in advancing fashion AI research and continues to be one of the most important resources for developing recommendation systems and computer vision applications in the fashion domain.

DeepFashion2 Dataset

DeepFashion2 is an advanced version of the original DeepFashion dataset and was developed to address several limitations found in earlier datasets. It provides more detailed annotations and additional information that support complex computer vision tasks.

The dataset includes clothing images captured from different viewpoints, lighting conditions, and poses. It also contains detailed garment landmarks, segmentation masks, bounding boxes, and category labels.

One of the key advantages of DeepFashion2 is its ability to support tasks such as pose estimation, clothing detection, segmentation, landmark localization, and virtual try-on research. The dataset is designed to represent real-world fashion scenarios more accurately than its predecessor.

DeepFashion2 has become an important benchmark dataset for researchers working on fashion recommendation systems and virtual try-on technologies. The additional annotations provide richer information that helps improve model accuracy and performance.

Due to its comprehensive structure and large-scale data collection, DeepFashion2 continues to be one of the most valuable datasets for fashion AI applications.

Importance of Virtual Try-On in E-Commerce

Virtual try-on technology has become increasingly important in modern e-commerce platforms because it helps bridge the gap between physical and online shopping experiences.

One of the primary challenges faced by online shoppers is the inability to physically examine or try products before purchasing them. This often leads to uncertainty regarding size, fit, appearance, and overall satisfaction.

Virtual try-on systems help solve these problems by allowing customers to preview garments digitally. This improves confidence during the decision-making process and creates a more engaging shopping experience.

For businesses, virtual try-on technology offers several advantages. It can reduce product return rates, improve customer trust, increase conversion rates, and enhance overall user satisfaction. These benefits contribute to improved business performance and customer retention.

As Artificial Intelligence continues to evolve, virtual try-on systems are expected to become more realistic, accurate, and widely adopted across fashion e-commerce platforms worldwide.

2D Virtual Try-On Findings

2D Virtual Try-On technology focuses on generating clothing images directly on a user's photograph. These systems use computer vision and deep learning techniques to digitally place garments onto a person's image while preserving important visual characteristics such as clothing texture, color, and style.

The primary objective of 2D virtual try-on systems is to create realistic outfit visualizations without requiring a complete three-dimensional representation of the user. This makes the technology faster, easier to implement, and more computationally efficient compared to 3D approaches.

During the research, it was observed that 2D virtual try-on models have significantly evolved over the years. Early models focused mainly on garment transfer, while modern architectures improve garment alignment, texture preservation, and image realism.

One of the major advantages of 2D virtual try-on systems is their efficiency. Since they work directly on images, they require fewer computational resources and can generate results relatively quickly. This makes them highly suitable for large-scale e-commerce platforms where fast response times are important.

Another advantage is ease of deployment. Most online shopping platforms can integrate 2D virtual try-on solutions without requiring specialized hardware or complex user inputs.

However, 2D virtual try-on systems also have limitations. Since they operate on two-dimensional images, accurately representing body shape and garment fitting can be challenging. The generated results may not always reflect how a garment would fit in real-world conditions.

Additionally, users can only view the outfit from a limited perspective, reducing the overall realism of the experience. Complex body poses, loose garments, and overlapping clothing items can further impact output quality.

Despite these challenges, 2D virtual try-on remains one of the most practical and widely adopted solutions in the fashion industry due to its efficiency, scalability, and lower implementation cost.

3D Virtual Try-On Findings

3D Virtual Try-On technology represents a more advanced approach to digital garment visualization. Unlike 2D systems, 3D virtual try-on creates a three-dimensional representation of both the user and the garment to simulate realistic clothing fitting.

The research showed that 3D systems provide a significantly higher level of realism and accuracy. By utilizing body models, garment models, and simulation techniques, these systems can generate realistic visualizations from multiple viewing angles.

One of the biggest strengths of 3D virtual try-on is its ability to accurately represent body shape and garment draping. This allows users to better understand how a clothing item may fit and appear in real-world situations.

Another advantage is enhanced interactivity. Users can often rotate virtual avatars, view garments from different perspectives, and explore outfit details more effectively than with traditional image-based methods.

3D systems are particularly valuable for custom clothing applications, luxury fashion, and premium shopping experiences where accurate fitting is essential.

However, these benefits come with additional complexity. Building accurate body models and garment simulations requires larger datasets, advanced algorithms, and significant computational resources.

Implementation costs are also higher compared to 2D approaches, making large-scale deployment more challenging for some businesses.

Despite these challenges, 3D virtual try-on technology is widely considered the future of digital fashion experiences due to its superior realism and user engagement capabilities.

VITON (Virtual Try-On Network)

VITON is one of the earliest and most influential virtual try-on models developed for image-based garment transfer. It introduced the concept of digitally fitting garments onto a person's image while preserving clothing appearance and body structure.

The model uses a coarse-to-fine approach to align garments and generate realistic outputs. It combines information from the person's image and the target garment to produce a final try-on result.

VITON played an important role in establishing the foundation for future virtual try-on research. Although newer models have improved upon its limitations, VITON remains a significant milestone in the development of virtual try-on technology.

The main advantage of VITON is its simplicity and effectiveness in demonstrating image-based garment transfer. However, the model may struggle with complex poses and fine garment details when compared to more advanced approaches.

CP-VTON

CP-VTON, which stands for Characteristic-Preserving Virtual Try-On Network, was developed as an improvement over the original VITON framework.

One of the major contributions of CP-VTON is its improved clothing alignment mechanism. The model utilizes advanced warping techniques to better fit garments onto the target person while preserving important clothing characteristics.

The research showed that CP-VTON generates more realistic outputs than VITON, particularly in terms of garment positioning and visual consistency.

The model significantly improved the handling of clothing deformation and garment fitting, making it one of the most widely referenced virtual try-on architectures in fashion AI research.

Although CP-VTON improved overall output quality, certain challenges related to image realism and texture preservation still remained.

HR-VITON

HR-VITON was developed to address limitations associated with image resolution and garment detail preservation.

One of the key objectives of HR-VITON is to generate high-resolution virtual try-on images while maintaining the visual quality of garments. The model places greater emphasis on preserving fine textures, patterns, and clothing details.

Compared to earlier architectures, HR-VITON produces sharper and more visually appealing outputs. The model improves the realism of generated images and reduces artifacts that may appear during garment transfer.

The research highlighted that HR-VITON achieves a good balance between realism and computational efficiency, making it a strong candidate for practical virtual try-on applications.

Its ability to preserve garment details makes it particularly useful for fashion products where texture and design patterns are important.

StableVITON

StableVITON represents one of the latest advancements in virtual try-on technology. The model leverages diffusion-based image generation techniques to create highly realistic try-on results.

Diffusion models have gained popularity in AI image generation because of their ability to generate detailed and visually appealing images. StableVITON applies these techniques specifically to virtual try-on tasks.

The model demonstrates excellent garment preservation, realistic fitting, and improved visual consistency. Compared to earlier approaches, StableVITON generates outputs that more closely resemble real photographs.

During the research, StableVITON consistently demonstrated some of the highest-quality results among the models studied. Its ability to maintain garment textures, colors, and fine details contributes significantly to output realism.

The model represents a major step forward in creating practical and visually convincing virtual try-on experiences.

IDM-VTON

IDM-VTON is another advanced virtual try-on model that focuses on generating realistic clothing visualizations while preserving garment details and body structure.

The model incorporates modern image generation techniques to improve fitting accuracy and visual quality. It addresses several limitations observed in earlier virtual try-on architectures.

One of the major strengths of IDM-VTON is its ability to generate realistic results while maintaining important garment characteristics such as texture, patterns, and design elements.

The research showed that IDM-VTON performs exceptionally well in complex clothing scenarios and demonstrates strong potential for future commercial applications.

Its advanced architecture and realistic outputs make it one of the most promising virtual try-on models currently available.

Model Comparison Report

The virtual try-on models explored during this research differ in terms of architecture, realism, garment preservation, and computational complexity.

VITON introduced the foundation of image-based virtual try-on technology and demonstrated the feasibility of digital garment transfer.

CP-VTON improved garment alignment and fitting accuracy through enhanced warping mechanisms.

HR-VITON focused on generating higher-resolution outputs and preserving clothing details.

StableVITON leveraged diffusion-based image generation to produce highly realistic virtual try-on results.

IDM-VTON further improved realism and garment preservation while addressing limitations observed in previous approaches.

Based on the research findings, StableVITON and IDM-VTON produced the most visually realistic outputs and demonstrated the strongest potential for future fashion applications. However, each model contributed important advancements that helped shape the evolution of modern virtual try-on technology.

Proposed AI Recommendation Workflow

Based on the research conducted during this internship, a personalized AI recommendation workflow was designed to demonstrate how recommendation systems can be integrated into fashion applications.

The workflow begins with collecting user information such as preferred clothing styles, favorite colors, fashion interests, shopping history, and product interactions. This information is used to build a personalized fashion profile for each user.

The recommendation engine analyzes the collected information using recommendation algorithms and generates personalized outfit suggestions. These recommendations are based on user preferences, product similarity, and behavioral patterns.

As users interact with the recommendations by viewing products, liking items, or making purchases, additional data is collected. This feedback is then used to continuously improve recommendation quality and personalize future suggestions.

The proposed workflow consists of the following stages:

User Registration

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Preference Collection

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Fashion Profile Creation

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Recommendation Engine

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Personalized Outfit Suggestions

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User Feedback

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Recommendation Improvement

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Better Future Recommendations

This workflow demonstrates how Artificial Intelligence can continuously learn from user behavior and deliver more accurate recommendations over time.

Challenges Faced

Throughout the internship, several challenges were encountered while conducting research on recommendation systems and virtual try-on technologies.

One of the primary challenges was understanding the architecture of advanced virtual try-on models. Many research papers contained complex deep learning concepts, technical terminology, and detailed mathematical explanations that required additional study and analysis.

Another challenge involved comparing different virtual try-on models. Since each model was developed using different datasets, architectures, and evaluation methods, direct comparison was not always straightforward. Careful analysis was required to understand the strengths and limitations of each approach.

Managing a large amount of research material also proved challenging. Multiple research papers, datasets, model architectures, and technical resources needed to be organized and reviewed systematically.

Additionally, understanding the practical applications of recommendation systems required studying both theoretical concepts and real-world implementations used by fashion platforms.

These challenges were addressed through structured research, documentation, comparison tables, and continuous learning throughout the internship period.

Key Learnings

This internship provided valuable exposure to Artificial Intelligence applications in the fashion industry and significantly improved my understanding of several important concepts.

One of the key learnings was understanding how recommendation systems personalize user experiences. Studying Content-Based Filtering, Collaborative Filtering, Hybrid Recommendation Systems, and Deep Learning approaches helped me understand how modern platforms generate personalized recommendations.

The internship also enhanced my knowledge of Computer Vision and Deep Learning technologies. Through the study of virtual try-on models, I learned how AI systems digitally fit garments onto users and generate realistic clothing visualizations.

Another important learning experience involved exploring fashion datasets. Understanding how datasets such as DeepFashion, DeepFashion2, FashionGen, and DressCode are used to train AI models provided insight into the importance of high-quality data in machine learning applications.

The project also improved my ability to analyze research papers, compare different AI models, organize technical information, and create structured documentation.

Overall, the internship strengthened both my technical knowledge and research skills while providing practical exposure to emerging AI technologies.

Future Improvements

Fashion AI technologies continue to evolve rapidly, creating opportunities for future innovation and development.

One potential improvement is the introduction of real-time virtual try-on systems that allow users to instantly visualize garments using live camera input. Such systems could provide more interactive and engaging shopping experiences.

Another area of improvement involves accurate body measurement estimation. Future AI models may be able to generate highly personalized fitting recommendations based on detailed body measurements and shape analysis.

The integration of Augmented Reality (AR) technology with virtual try-on systems could further improve realism and user interaction. AR-based solutions would allow customers to view garments in real-world environments before making purchasing decisions.

Recommendation systems can also become more intelligent by incorporating social media trends, customer preferences, seasonal factors, and fashion influences into recommendation generation.

Advanced diffusion models and generative AI technologies are expected to further improve image quality, realism, and garment preservation in virtual try-on applications.

In the future, the combination of recommendation systems, virtual try-on technology, computer vision, and generative AI may create highly personalized and immersive fashion shopping experiences.

Conclusion

The completion of this project provided valuable exposure to the practical applications of Artificial Intelligence in the fashion industry. Throughout the internship, I researched recommendation systems, virtual try-on technologies, fashion datasets, and AI workflows to understand how these technologies contribute to improving online shopping experiences.

The study of recommendation systems helped me understand how platforms analyze user preferences and behavior to provide personalized product suggestions. Exploring Content-Based Filtering, Collaborative Filtering, Hybrid Recommendation Systems, and Deep Learning approaches gave me insight into the techniques used to enhance customer engagement and satisfaction.

Research on virtual try-on technologies allowed me to understand how Artificial Intelligence and Computer Vision can be used to digitally fit garments onto a user's image. By studying models such as VITON, CP-VTON, HR-VITON, StableVITON, and IDM-VTON, I learned about the advancements that have been made in improving garment alignment, image quality, realism, and overall user experience.

One of the key findings of this research is that the combination of recommendation systems and virtual try-on technologies has the potential to significantly transform the future of online fashion retail. Personalized recommendations can help users discover products that match their preferences, while virtual try-on systems can reduce uncertainty by allowing customers to visualize garments before making a purchase.

In addition to technical knowledge, this internship improved my research, analytical thinking, problem-solving, and technical documentation skills. The experience provided practical exposure to emerging AI technologies and strengthened my understanding of how AI solutions are developed and applied in real-world scenarios.

Overall, this internship has been a valuable learning experience that enhanced my understanding of Artificial Intelligence, Deep Learning, Computer Vision, and recommendation systems while providing insight into the future of fashion technology and e-commerce.

